

OPERA: Open-World Performance Evaluation for Biomedical Research Agents

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Abstract

We present OPERA (Open-World Performance Evaluation for Biomedical Research Agents), an evaluation framework for long-horizon, open-ended biomedical discovery tasks. Unlike closed-form benchmarks that score against fixed answers, OPERA captures the agent’s full reasoning trace—hypothesis evolution, evidence accumulation, and tool usage—and evaluates the *quality of scientific process* rather than final correctness. The framework introduces a three-persona pipeline (Task Proposer, Evaluator, Reviewer) with structured artifact handoffs, a rubric-driven LLM-as-judge scorer with multi-backend consensus checking, and a novel martingale score that quantifies whether hypothesis confidence updates are evidence-driven. We demonstrate OPERA on two pilot biomedical tasks using both Claude and Gemini agents and argue that open-world process evaluation is necessary for benchmarking agents on tasks where ground truth is unknown.

1 Introduction

Large language model (LLM) agents are increasingly deployed on scientific discovery tasks: literature mining, hypothesis generation, experimental design, and data analysis [3, 5]. Yet existing benchmarks measure whether an agent reaches a known answer—a regime that fundamentally mismatches the structure of real research, where ground truth is unknown, tasks span weeks of effort, and quality is a property of the *reasoning process*, not just the conclusion.

We identify three specific gaps:

1. **Fixed-answer evaluation** cannot score open-ended tasks whose correct answer is not yet known to humanity.
2. **Single-turn or short-horizon tasks** do not capture the multi-step hypothesis revision that characterizes real discovery.
3. **Monolithic scoring** conflates tool-use breadth, evidence quality, and novelty—obscuring which agent capabilities matter.

OPERA (Open-World Performance Evaluation for Biomedical Research Agents) addresses all three by providing: (1) a task specification format that encodes *evaluation criteria* rather

than expected outputs, (2) a structured hypothesis ledger that records the full belief-update trajectory, and (3) a multi-stage reviewer pipeline that combines automated metrics with rubric-guided LLM judging and optional human expert review.

Concretely: in our pilot SLE Biomarker Discovery task (§4), an agent is given a public single-cell RNA-seq cohort and asked to propose candidate cell-type-specific biomarkers with multi-modal supporting evidence—a task whose correct answer is not known until the candidates are validated in follow-up studies, and which the agent may pursue via any combination of statistical analysis, literature search, database queries, and hypothesis revision it chooses. We use this task to ground the framework description that follows; §4 gives the full task specification and rubric.

2 Related Work

Open-world evaluation. Kapoor et al. [6] distinguish a spectrum of evaluation approaches from Q&A benchmarks and outcome-only agent benchmarks through to *open-world evaluations*: long-horizon tasks in real-world settings where success cannot be automatically graded and qualitative log analysis is necessary. They argue that as AI capabilities grow, evaluations must grow correspondingly complex, and that open-world evaluations provide early warning of frontier capabilities that benchmarks cannot surface. OPERA is an instance of this class, applied to biomedical discovery. Where Kapoor et al. focus on characterizing open-world evaluations as a methodology, OPERA contributes concrete infrastructure and metrics—a structured hypothesis ledger, process-level automated scores, and a rubric-driven multi-stage reviewer—that make open-world scientific evaluations reproducible and extensible.

Biomedical agent benchmarks. A cluster of recent benchmarks evaluates LLM agents on biological and biomedical analysis tasks, but all share a common assumption: a correct answer is known at evaluation time. BioMysteryBench [1] presents 99 professional-level bioinformatics problems paired with verifiable answer rubrics derived from data properties; BixBench [9] evaluates agents on 61 expert-curated computational biology capsules scored by binary LLM judgment; and BioML-Bench frames biological machine learning as a Kaggle-style competition with deterministic metrics against ground truth. AstaBench [13], the broadest of these, spans 2,400+ problems across the full research pipeline—literature understanding, code execution, data analysis, and end-to-end discovery—but its E2E evaluation still requires a defined outcome against which agent outputs are scored. More open-ended systems such as AutoDiscovery [8] and AI Scientist [7] automate hypothesis generation and testing but still rely on closed verification—a fixed dataset or executable experiment—to score outcomes. What none of these frameworks can measure is the quality of an agent’s reasoning *process* on a question whose answer is not yet known. When the task itself is a genuine open scientific question—what biomarkers predict lupus flare in an understudied cohort?—there is no key to grade against. OPERA is designed precisely for this setting.

Belief update metrics and Bayesian trajectory scoring. A key insight motivating OPERA’s process metrics is that LLMs systematically fail to update beliefs correctly in individual sequential reasoning chains, even when their average behavior across independent queries approximates Bayesian rationality [11]. This failure manifests as belief entrenchment: whatever confidence an agent currently holds tends to be reinforced by subsequent reasoning steps, regardless of evidence. The Martingale Score [14] formalizes this as the Pearson correlation

between b_t and Δb_t —a statistic that can be computed without ground truth and that has been shown to correlate with accuracy on domains where labels are available. OPERA adopts this metric directly, applying it to the confidence sequences recorded in the hypothesis ledger. For post-hoc hypothesis validation, OPERA draws on POPPER [12], which applies sequential falsification via e-value accumulation—rather than p-values, which require a pre-specified stopping time—to assess whether accumulated evidence clears a statistically rigorous bar. The e-value framework’s validity at any stopping time is essential in our variable-length agent runs. The practical implication of [11] for OPERA’s design is the requirement that the falsification observer be a separate model instance: asking the running agent to self-assess its evidence is unreliable precisely because the same model that generated the beliefs will entrench them.

LLM-as-judge scoring. Zheng et al. [15] establish that language models can reliably assess open-ended outputs against rubric criteria, forming the basis for our LLM-as-judge scoring stage. OPERA extends this in two directions: the two-pass evidence-blinded protocol ensures scores reflect evidence quality rather than rhetorical fluency, and running multiple judge back-ends in parallel converts inter-judge disagreement from noise into a signal—contested criteria are surfaced to the human reviewer rather than resolved by averaging. Austin et al. [2] show that evaluating both execution *and* explanation of code reveals reasoning that output-only grading misses. OPERA applies the same principle to scientific reasoning: an agent must not only produce hypotheses but maintain a traceable, evidence-anchored belief trajectory that can be audited independently of its stated rationale.

3 The OPERA Framework

OPERA structures each evaluation run as a pipeline of three well-separated roles—Task Proposer, Evaluator, and Reviewer—each with defined input and output contracts (Figure 1). Separation of concerns is a deliberate design choice: the agent under evaluation never has access to the rubric used to score it, and the scoring stage never communicates with the agent directly. This prevents the two most common failure modes in LLM-based evaluation: teaching to the test and post-hoc rationalization.

3.1 Task Specification and the Rubric Contract

A central design choice in OPERA is that evaluation criteria are specified independently of expected answers. Each task is defined by two artifacts produced by the Task Proposer: a machine-readable task specification encoding the scientific goal, termination conditions, and a structured rubric; and an agent-facing task prompt written in purely goal-oriented language that describes what a successful outcome looks like without prescribing methodology.

The rubric is the single source of truth for what constitutes quality in a run. Each rubric item carries a criterion identifier, a descriptive label, a scope (per-hypothesis or run-level), a numeric scale, and a *guidance* string written for a domain scientist. Critically, the same guidance text is injected verbatim into LLM scoring prompts and rendered in the human reviewer’s interface—preventing criterion drift between automated and human evaluation and making the evaluation criteria fully auditable. An LLM-assisted authoring tool lowers the barrier to task creation by scaffolding both artifacts from a free-text description, though hand-authoring is equally supported.

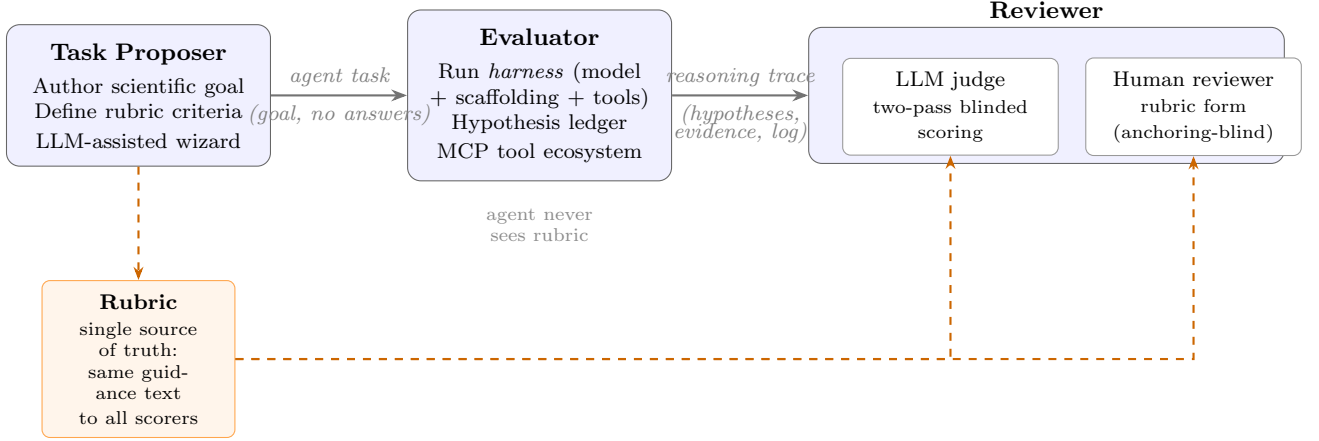


Figure 1: OPERA’s three-persona pipeline. Solid arrows carry run-time artifacts; dashed orange arrows show the *rubric*—authored once by the Task Proposer and injected verbatim into both the LLM judge and the human reviewer’s scoring form. This shared contract prevents criterion drift between automated and human evaluation. The agent under evaluation never has access to the rubric.

3.2 Agent Execution and the Hypothesis Ledger

OPERA evaluates a *harness*—the full system comprising model, scaffolding, tool access, and any agent-specific prompting strategy—not an isolated model. Two runs using the same underlying model but different harnesses may produce meaningfully different belief trajectories, and OPERA’s metrics are designed to surface exactly that variation.

The Evaluator stage currently supports two widely-used agent harnesses—Claude Code and Gemini CLI—connected to a shared tool ecosystem via the Model Context Protocol (MCP). Because the Evaluator’s only requirement is that the agent record structured artifacts (hypotheses, evidence, run log) through defined interfaces, any MCP-compatible harness can participate without changes to the evaluation infrastructure. This cleanly separates harness-specific behavior from hypothesis tracking and scoring, which remain uniform across agent systems.

Agents operate with broad tool access by design: harness-native capabilities (shell execution, web search, file I/O) are unrestricted, and two MCP layers add structured instrumentation. The hypothesis-ledger MCP provides six operations for managing hypothesis state: recording a new hypothesis with an initial confidence score, updating confidence with supporting evidence and a rationale, and marking a hypothesis terminal (confirmed or refuted). A key constraint is the enforced separation of evidence from rationale—evidence entries must be factual (observed measurements, database identifiers, statistical values, literature citations) while rationale entries are interpretive, explaining why confidence changed. This separation is a prerequisite for the two-pass scoring design described below. A second MCP layer provides access to over 600 pre-built biomedical tools spanning literature retrieval, genomics databases, pathway analysis, and drug-target interaction [4], enabling agents to pursue multi-modal evidence without bespoke integration work. The agent is free to use any combination of these capabilities; which tools it reaches for is itself a behavioral signal captured in the tool-diversity metric.

3.3 Multi-Stage Review and Scoring

The Reviewer stage operates entirely offline on the artifacts produced by the Evaluator and is designed to yield three complementary signals: automated process metrics, an independent LLM-based assessment, and optional human expert adjudication.

Automated metrics quantify the structural properties of the agent’s reasoning trajectory without any LLM calls. Four metrics characterize hypothesis testing behavior in context: the number of turns used (out of the allocated budget), the total number of hypotheses formed, the mean number of belief revision events per hypothesis (i.e., how many times the agent returned to update a hypothesis after new evidence), and the refutation rate (the fraction of hypotheses explicitly rejected). Reporting turns alongside hypothesis counts is essential for interpretation: an agent that formed five hypotheses in ten turns is meaningfully different from one that formed five hypotheses in 250 turns. Zero revisions on a hypothesis is itself a signal—it indicates the agent registered a candidate but never re-examined it in light of subsequent evidence. Together these metrics characterize whether the agent engaged in genuine iterative reasoning or merely accumulated hypotheses without revisiting them.

Tool breadth is captured separately for each tool layer, using the normalized Shannon entropy $\hat{H}$:

$$\hat{H} = \frac{-\sum_i p_i \log_2 p_i}{\log_2 N}, \quad (1)$$

where p_i is the fraction of calls to tool i and N is the number of tools available in that layer. Normalizing by $\log_2 N$ yields a value in $[0, 1]$ regardless of layer size, making the two layers—the six hypothesis-management operations and the 600+ biomedical tools—directly comparable. A score near 1 indicates the agent explored the available tool space broadly; a low score indicates concentration on a small subset. Neither extreme is normatively better: in real research scenarios, an agent may legitimately rely on a few high-value tools repeatedly, so low $\hat{H}$ reflects research strategy as much as breadth.

A belief entrenchment indicator is computed as the Pearson correlation r between a hypothesis’s current confidence b_t and its subsequent one-step change $\Delta b_t = b_{t+1} - b_t$, pooled across all hypotheses in the run [14]:

$$r = \text{Pearson}(b_t, \Delta b_t). \quad (2)$$

A near-zero r indicates updates are unpredictable from the current confidence level, consistent with evidence-driven revision; a positive r signals entrenchment, where already-high confidence tends to keep rising regardless of new evidence.

The confidence values b_t are the agent’s *expressed* beliefs, reported through the hypothesis-ledger interface. They are not independently verified—an agent could in principle report arbitrary values. Two design choices bound this concern. First, the agent never has access to the rubric, so there is no scoring target to game confidence values toward. Second, the falsification observer independently derives a confidence estimate from the evidence trail alone and compares it to the agent’s self-reported values (Eq. 3); systematic divergence between the two is a direct signal of overconfidence or anchoring. The martingale score therefore measures internal self-consistency of expressed beliefs, while the observer comparison measures their external grounding.

Because the statistic is pooled across all $(b_t, \Delta b_t)$ pairs in the run, its reliability depends directly on the total number of revision events: runs with few revisions yield a highly variable estimate and should be treated as descriptive rather than conclusive. All automated metrics are computed offline from the recorded reasoning trace and included in the review report.

LLM-as-judge scoring uses a two-pass protocol designed to isolate evidence quality from agent rhetoric. In the first pass, the judge model receives only the factual evidence trail for each hypothesis—tool outputs, citations, quantitative findings—and scores each rubric criterion without access to the agent’s stated reasoning. In the second pass, the judge is shown the agent’s own rationale and asked to characterize agreement or divergence, but cannot revise the first-pass scores. This blinding ensures that a well-articulated but evidence-poor argument cannot inflate criterion scores. To reduce dependence on any single judge model, all configured backends run in parallel (in the pilot: Claude Haiku 4.5 and Gemini 2.5 Flash); when two or more backends score the same criterion and their scores disagree by two or more points on a five-point scale, the criterion is flagged as *contested* and surfaced prominently to the human reviewer.

An additional falsification observer provides an external check on the agent’s expressed confidence. A separate LLM instance (Claude Haiku 4.5) reads only the accumulated evidence trail for each hypothesis—never the agent’s stated confidence or rationale—and independently estimates evidential support via e-value accumulation, following the sequential falsification framework of POPPER [12]. Individual e-values e_i are derived from p-values and effect sizes cited in the evidence; the product $E = \prod_i e_i$ is mapped to a calibrated confidence estimate:

$$C = \frac{E}{1 + E}, \quad E = \prod_i e_i. \quad (3)$$

Comparing C to the agent’s self-reported b_t provides a direct measure of overconfidence or anchoring. Crucially, the observer’s validity is independent of whether the agent formally concluded a hypothesis: any hypothesis for which evidence has been recorded is assessable, and an agent that accumulates strong evidence without updating its stated confidence is equally detectable as one that explicitly concludes. This means the falsification check runs on the full set of non-rejected hypotheses at run end, not only those the agent chose to formally declare.

Finally, a review package is assembled for a domain scientist, presenting the task context, rubric, and full step-by-step agent log. LLM scores are concealed until after the human submits their own rubric ratings, preventing score anchoring. This design allows independent human adjudication without requiring access to model APIs or evaluation infrastructure.

4 Pilot Tasks and Preliminary Results

We instantiate OPERA on two biomedical discovery tasks to demonstrate the framework’s generality and to report initial automated metrics.

4.1 Task Instantiations

The first task, *SLE Biomarker Discovery*, asks an agent to identify novel candidate biomarkers for systemic lupus erythematosus (SLE) disease activity and flare prediction. The agent is provided a starting reference dataset—a 1.2M-cell single-cell RNA-seq PBMC cohort [10]—but is otherwise unconstrained: it may access any publicly available databases, literature, protein interaction networks, GWAS catalogs, drug-target repositories, and computational tools it can discover through the tool ecosystem. The rubric evaluates five dimensions: literature grounding of proposed candidates, quality of multi-modal evidence integration, novelty relative to the established SLE literature, statistical rigor of supporting evidence, and feasibility of proposed clinical validation pathways.

The second task, *Protein Synergy Discovery*, asks an agent to identify pairs or networks of proteins with synergistic therapeutic potential in a specified disease context, drawing on interaction databases, expression data, and mechanistic literature. The rubric assesses mechanistic plausibility of proposed synergies, evidence breadth across modalities, diversity of hypotheses explored, and the quality of proposed experimental validation designs. Together these tasks represent two distinct scientific reasoning regimes: a data-anchored biomarker identification task with an existing reference dataset, and an open-ended network-level discovery task driven primarily by literature and interaction databases.

4.2 Results

We report automated and LLM-judge metrics from three pilot runs (30-turn budget) on the SLE Biomarker Discovery task, using two frontier model families under identical task specifications (Table 1). For open-ended discovery tasks there is no predetermined correct turn count; 30 turns is enough to form and revise hypotheses, but more compute allows deeper literature integration, tool exploration, and iterative refinement. Longer runs and human expert adjudication are ongoing; the results here establish that OPERA’s evaluation layers produce meaningful signal even at modest scale.

Table 1: Automated process metrics for pilot runs on the SLE Biomarker Discovery task. Runs A and B are two *independent* sessions of the same model (Claude Sonnet 4.6) on the identical task and prompt, illustrating genuine run-to-run variability rather than different configurations; Run C uses a different model family (Gemini 2.5 Pro). Turns: model invocations used; Hyp.: total hypotheses formed; Rev./Hyp.: mean revision events per hypothesis; Refuted: fraction explicitly rejected; Martingale r : belief entrenchment indicator (pooled Pearson correlation; n = total revision pairs).

Model	Run	Turns	Hyp.	Rev./Hyp.	Refuted	Martingale r
Claude Sonnet 4.6	A	31	5	0.4	20%	N/A ($n=2$) [†]
Claude Sonnet 4.6	B	31	3	2.3	0%	-0.43 ($n=7$)
Gemini 2.5 Pro	C	30 [‡]	3	1.0	0%	0.00 ($n=3$)

[†] $n < 3$ revision pairs; score undefined.

[‡] Budget-equivalent; Gemini CLI records no per-turn counter.

Automated metrics reveal qualitatively distinct search strategies. Across three runs at the same 30-turn budget, the agents produce strikingly different hypothesis trajectories (Table 2; Fig. 2 details Run B). Run A (Claude) spread across five hypotheses with shallow per-hypothesis investigation (0.4 revisions each) but explicitly refuted one candidate (20% refutation rate), indicating an active breadth-first falsification strategy—and used Bash-based scripting as its primary tool (69 shell invocations). Run B (Claude) converged on three candidates with deep iterative revision (2.3 revisions each) and no formal refutations—a concentrated, literature-driven style (WebSearch $\times 15$). Run C (Gemini) fell between: three hypotheses, moderate revision depth, symmetric belief updating ($r = 0.00$). These behavioral profiles—including tool-use signatures that distinguish shell-heavy from search-heavy exploration—are invisible to output-only evaluation; they are precisely what the structured hypothesis ledger exists to capture. The martingale estimates ($n \leq 7$) are too few for reliable conclusions at this scale, but they are directionally consistent with the revision profiles: Run B’s $r = -0.43$ reflects

predominantly downward confidence revisions, consistent with a run where updates tracked contradictory evidence (e.g., SIGLEC1’s non-specificity across interferonopathies).

Multi-layer evaluation surfaces measurement–reasoning gaps. Run B’s 0% refutation rate would, taken alone, suggest the agent never challenged its hypotheses—a flag that would cause a single-metric evaluation to dismiss the run as uncritical. OPERA’s LLM-as-judge layer, scored across two independent backends, overturned this reading: the agent explicitly acknowledged SIGLEC1’s cross-reactivity with COVID-19 interferonopathies, flagged absent GWAS support for GZMH, and distinguished diagnostic from therapeutic utility for each candidate. The zero formal refutation rate was attributed to a *reporting gap, not a reasoning gap*: adversarial reasoning was present in the confidence revisions, but the agent never translated it into an explicit status update. The judge also caught a subtler failure in the same run: for GZMH, the agent *increased* its expressed confidence from 0.68 to 0.70 after noting four explicit evidence gaps—behavior invisible to output-only evaluation but directly readable from the ledger’s revision trail.

Falsification observer reveals model-level calibration differences. Comparing agent-expressed confidence to the independent observer’s evidence-derived confidence reveals a consistent pattern across harnesses. Both Claude runs showed observer confidence at or above the agent’s self-report: Run A averaged a positive delta across four hypotheses (IFI44L -0.03 , TNFSF13B $+0.12$, LILRA4 $+0.17$), and Run B was positive for all three ($+0.07$, $+0.08$, $+0.01$)—indicating that Claude agents tend to be conservative or well-calibrated relative to evidence strength. Run C reversed the pattern: all three Gemini hypotheses showed negative deltas (-0.08 , -0.18 , -0.08), consistent with the flat martingale ($r = 0.00$) and lower revision depth—the agent expressed uniformly high confidence regardless of evidence weight. Neither the task output nor any single-layer metric would surface this difference; it emerges only through the pairing of structured belief records with an independent evidence assessment.

No evidence of explicit judge self-preference; implicit recognition untested. Because both the evaluated agents and the LLM judges are drawn from the same two model families (Claude, Gemini), self-preference—a judge favoring output from its own family—is a standing risk of any LLM-as-judge design. The judge prompt contains no model or harness identifier, only the hypothesis, evidence trail, and agent rationale, which rules out *explicit* identity-based self-preference by construction but leaves *implicit* stylistic self-recognition untested—judges have been shown to recognize and favor their own outputs from writing style alone, without being told the source. Testing the outcome-level version, pooling every numeric criterion score across the three pilot runs and splitting by whether the judge backend shares a model family with the run’s agent (same-family) or not (cross-family), same-family scores average *lower* than cross-family scores (3.23 vs. 3.51, $n = 39$ and 35 scale-rated criteria)—the opposite of what self-preference would predict. A more consistent pattern instead: the Gemini Flash judge scored higher than the Claude Haiku judge in all three runs, including on Claude-generated work (Runs A, B) and relative to Claude Haiku judging Gemini’s own work (Run C)—a general leniency asymmetry between judge backends rather than a preference tied to output source, though implicit style recognition cannot be fully excluded with only three runs to disentangle it from calibration. As with the martingale score, this check is too small ($n \leq 15$ per cell) to be conclusive, but it is a judge-bias question invisible to single-backend evaluation and one OPERA’s multi-backend design makes directly checkable.

Table 2: Per-hypothesis summary across all three pilot runs. Rev.: confidence-updating revisions. Evid.: recorded evidence items. b_T : final agent-expressed confidence (R=refuted). $\hat{C}$: independent observer confidence derived from evidence trail only. Recurs.: runs in which the gene was independently nominated.

Run	Gene	Cell type	Rev.	Evid.	b_T	$\hat{C}$	Recurs.
A	IFI44L	Mono./pDC	0	3	0.92	0.89	3/3
	SIGLEC1	Mono.	1	3	0.75	—*	2/3
	TNFSF13B	Mono.	0	3	0.80	0.92	1/3
	LILRA4	pDC	0	2	0.55	0.72	1/3
	IKZF2	—	1	2	R	—	—
B	IFI44L	Mono./pDC	2	5	0.88	0.89	3/3
	SIGLEC1	Mono.	2	5	0.80	0.87	2/3
	GZMH	CD8 ⁺ T	3	4	0.70	0.78	1/3
C	IFI44L	B cell	1	3	0.80	0.72	3/3
	ISG15	B cell	1	3	0.80	0.62	1/3
	MX1	B cell	1	3	0.80	0.72	1/3

* No quantitative p-values in evidence; observer returned uninformative prior.

Cross-run convergence on IFI44L. IFI44L emerged as the top-ranked hypothesis in all three runs independently: Runs A and B (Claude) and Run C (Gemini) each assigned it the highest agent confidence, using different tool strategies and starting from different points in their respective searches. In Run B it was supported by GWAS hits, causal mouse knockout evidence, and a mechanistic IFN- α /c-Jun/AP-1 axis (confidence 0.88; observer 0.89). SIGLEC1 appeared in both Claude runs and scored 5/5 on evidence strength and clinical relevance from the independent LLM judge. Without a persistent, structured hypothesis record across runs, this agreement would be invisible—each run’s output is a standalone document, and manual comparison across three unstructured reports would not reliably surface the convergence signal. Human expert adjudication is ongoing.

5 Discussion and Limitations

Human expert availability. The human review step requires a domain scientist, which limits throughput. We mitigate this by surfacing only contested LLM-scored criteria for human adjudication, reducing the expert’s workload to the cases where automated consensus breaks down.

Rubric design lessons from the first human review. Piloting the review process with a domain expert (an SLE researcher, scoring Run B) surfaced lessons for rubric-driven evaluation that are not specific to OPERA’s biomedical setting. Most consequential: rubric criteria specified a scale and a description of what to weigh, but no anchors for what evidence should earn each score—reviewers, human or LLM, calibrated against no shared reference point. Behaviorally-anchored criteria, with example evidence illustrating a low versus a high score, together with a task-level floor scenario showing a minimal-effort baseline run, would let human and LLM judges converge on the same scale rather than each inventing their own. Two further risks are structural to rubric-driven evaluation generally, independent of who is scoring: *evidence locality*, since a reviewer cannot be assumed to retain and correctly re-apply an entire

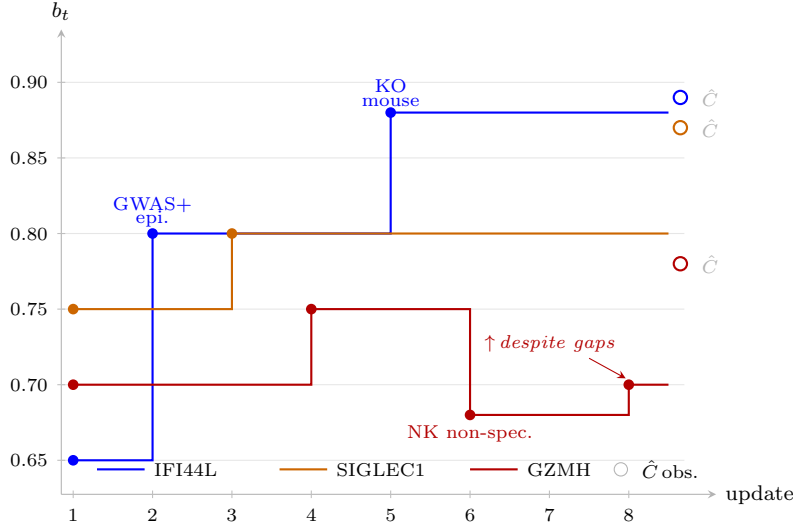


Figure 2: Belief trajectories for Run B (Claude Sonnet 4.6). Each step reflects a confidence-updating ledger revision; events are numbered in chronological order. All three hypotheses begin within seconds and diverge as evidence accumulates. Hollow circles ($\hat{C}$) show the independent observer’s confidence computed from the evidence trail alone—all three sit above the agent’s final b_T , confirming appropriately conservative self-reporting. The GZMH trajectory (red) rises at event 8 despite the agent explicitly cataloguing four evidence gaps at event 6, a pattern detectable only in the revision trail.

evidence trail across separately elicited criterion scores; and *construct conflation*, when guidance bundles several logically separable questions under one criterion—our novelty criterion, for instance, asked simultaneously about prior-literature grounding, evidence integration, and clinical impact—yielding a single score whose meaning is ambiguous to interpret afterward. Because OPERA’s rubric is injected verbatim into both the LLM judge and the human form (§3), addressing any of these strengthens both evaluation layers at once.

Ground truth will eventually exist. For some tasks (e.g., proposed biomarkers that are later validated in independent studies), retrospective ground truth becomes available. OPERA’s rubric-based scoring can be compared against such retrospective labels as a long-term calibration signal.

Small-sample process metrics. The martingale score and the judge-backend calibration check in §4 both pool few data points per run ($n \leq 7$ revision pairs and $n \leq 15$ per judge/family cell, respectively). At this scale both are descriptive signals, not statistically powered tests; we report them because they are directionally consistent with independently observed behavior (revision depth, refutation rate) and because no single-layer metric would surface them at all, but neither should be read as a confirmed effect until validated on more runs.

Harness coverage and generalization. Because OPERA evaluates harnesses rather than models, its findings do not generalize directly to model capability: a result obtained with one

scaffolding strategy may not hold under a different prompting approach, tool configuration, or agent framework. Currently, Claude Code and Gemini CLI serve as reference harnesses; the MCP interface allows additional agent frameworks to be integrated. Systematic comparison across harnesses for the same underlying model is a natural next step and would help disentangle model contribution from scaffolding contribution in observed performance.

Applicability and future extensions. OPERA is best suited to discovery tasks that admit a structured hypothesis representation—a candidate, a claim, accumulating evidence, and a revisable confidence—rather than tasks whose output is a single artifact evaluated holistically (e.g., code generation, writing). Within that class, we expect the framework to generalize readily to other biomedical discovery problems (drug repurposing, pathway analysis, variant interpretation) and, with the ledger schema adapted to the domain’s evidence types, to non-biomedical open-world discovery more broadly. Two extensions we leave to future work: scaling the pilot evaluations across additional biomedical sub-domains (e.g., structural biology alongside translational genomics) to establish generalizability at greater breadth, and a sensitivity analysis of how minor prompt variations in the Evaluator persona affect multi-backend judge consensus. Both require substantially more pilot runs than the present study’s scale supports.

Preliminary results. This paper presents the OPERA framework and pilot task instantiations; full quantitative results with human expert scoring are ongoing. We release the codebase and task specifications to enable the community to reproduce and extend our evaluations.

6 Conclusion

OPERA provides a principled framework for evaluating LLM agents on open-world biomedical discovery tasks. By separating task specification, agent execution, and multi-modal review into three well-defined personas with structured artifact handoffs, and by introducing process-level metrics (martingale score, falsification observer, two-pass LLM judging), OPERA enables evaluation of scientific reasoning quality in the absence of ground truth. We argue this evaluation paradigm is necessary as AI systems are deployed on frontier scientific problems—and we release OPERA as an open framework for the community to extend.

Code and task specifications: <https://github.com/LanGuo/opera>

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